



Assignment/ Tutorial / Project / Report Cover Sheet

Programme : Bachelor of Software Engineering (Honours)

Subject Code : BTPR1013

Subject Name : Internet Applications

Lecturer Name : Ms Khairunnisha Ismail

Academic Session : 2025A

Detail:

Assignment / Tutorial / Project / Report Title:	Individual Assignment
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Due Date:	25 April 2025

Assessment:

MARK:	Comments:
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Rubrics

This project will be evaluated as follows:

Criteria	0-2 (Poor)	3-5 (Fair)	6-8 (Good)	9-10 (Excellent)	Marks
Understanding of Internet Applications	Minimal or unclear explanation, lacks relevance.	Basic discussion, lacks depth or real- world examples.	Well-explained with relevant industry examples.	In-depth analysis with strong insights and real-world relevance.	____/10
Discussion on Emerging Internet Trends	Trend is unclear or lacks connection to industry.	Basic explanation, minimal industry impact discussed.	Well-explained trend with relevant industry examples.	Excellent analysis with strong real-world impact and examples.	____/10
Webpage Content & Relevance	Content is unclear, lacks structure or relevance to the trend.	Basic content, lacks depth or connection to report.	Well-structured content explaining the trend with clarity.	Highly engaging, informative, and directly linked to the report topic.	____/10
Webpage Design & Technical Implementation	Poor layout, missing styling, or non-functional webpage.	Basic layout with minimal CSS, lacks consistency.	Well-structured layout with proper CSS styling and interactivity.	Visually appealing, responsive, and includes interactive elements.	____/10
Report Formatting & Integration	Poor formatting, lacks structure, or webpage not linked.	Basic formatting, lacks consistency, weak webpage connection.	Well-structured report with clear connection to the webpage.	Professionally formatted report, well-integrated webpage, excellent readability.	____/10

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Section 1: Introduction

Internet is a global network of interconnected computers and devices that allows users to access and share information and devices, allowing them to communicate and exchange data. It enables users to access a wide range of services, such as websites, emails, social media, online applications, and cloud storage.

Internet application is a broad term that refers to any application that connected to the internet for its operation. This can include applications that run on servers and are accessed through various protocols, not limited to HTTP/HTTPS. For example, email services like Gmail or VoIP (Voice over Internet Protocol) services as Skype are all considered Internet applications.

Section 2: Internet Applications in Different Fields

2.1 Healthcare

I. mHealth (Mobile Health apps) and Wearables

Mobile health apps are intended for the diagnosis, treatment, or prevention of health conditions or diseases. They can store and track health information, offer reminders, and use GPS tracking to direct or alert patients.

Wearable technologies fall within the realm of mobile health technologies and are specific add-on devices or sensors that connect with smart phones or watches, tablets, and cloud-based computing systems to collect the wearer's health data and health metrics.

- Benefits
 - Medication reminders
 - Symptom tracking and preventive care
 - Mental health support
- Real-world relevance
 - Apple Watch offers ECG and blood oxygen monitoring, with data that can be shared with physicians.

II. **Electronic Health Records (EHRs)**

Electronic Health Records (EHRs) are internet-based systems store and manage patient records securely, allowing healthcare providers to access and update information quickly. They provide real-time, patient-centered records that make information available instantly and securely to authorized users. EHRs facilitate better seamless data exchange between various providers, from primary care physicians to specialists and hospitals, ensuring integrated, patient-centric care.

- Benefits
 - Improve data sharing leads to fewer medical errors
 - Reduce duplication of tests
 - More holistic view of a patient's history especially patient's life-threatening allergy
- Real-world relevance
 - U.S. Department of Veterans Affairs (VA) uses a unified EHR system to provide seamless care across hundreds of facilities.

III. **Telemedicine and Telehealth**

Telehealth has become a cornerstone of modern healthcare, especially since the COVID-19 pandemic forced rapid adoption. It enables real-time consultation with healthcare providers through secure video platforms, which is particularly critical in rural or underserved areas where access to specialists is limited.

- Benefits
 - Reduce healthcare delivery costs
 - Improve chronic disease management
 - Enhance convenience for patients
- Real-world relevance
 - Sunway Medical Centre launched a Telemedicine Command Centre in 2021, providing 24/7 healthcare advisory services via phone, email, and social media platforms like WhatsApp and Facebook.

2.2 Transportation

I. Real-Time Navigation Apps

Real-time navigation apps leverage GPS, cloud computing, and real-time traffic data to provide users with optimal routes, estimated arrival times, and dynamic rerouting based on current road conditions.

- Benefits
 - Reduce travel time and fuel consumption
 - Dynamically reroute vehicles supports sustainability goals and reduces emissions.
- Real-world relevance
 - Waze is widely used in Malaysian cities such as Kuala Lumpur and Penang, it crowdsources traffic data from drivers to update users in real time about congestion, accidents, or road closures.

II. Ride-Sharing Platforms

Ride-sharing platforms connect passengers with drivers through mobile applications, utilizing GPS, real-time data analytics, and payment gateways to facilitate seamless transportation services. Ride platforms integrate mapping, payment processing, and rating systems online to create an ecosystem that not only optimizes individual rides but also reduces overall traffic congestion and parking demands in crowded city centers.

- Benefits
 - Provide gig economy jobs
 - Enhance flexibility and accessibility in areas with limited public transportation options.
- Real-world relevance
 - Grab offers services ranging from ride-hailing to food delivery and digital payments, adapting to local needs and preferences.

III. **Intelligent Transportation Information Systems (ITIS)**

Intelligent Transportation Systems (ITIS) combines hardware and software to collect, process, and analyze data from various sources, such as traffic signals, surveillance cameras, and vehicle sensors. The collected data is used to make real-time decisions or provide actionable insights that enhance traffic management and transportation planning.

- Benefits
 - Smart traffic management
 - Public transit optimization
 - Emergency response coordination
- Real-world relevance
 - ITIS operated by Dewan Bandaraya Kuala Lumpur (DBKL), ITIS uses CCTV cameras, traffic sensors, and GPS data to monitor road conditions and manage traffic across Greater Kuala Lumpur.

Section 3: Emerging Internet Trends in Transportation - Connected Autonomous Vehicles

3.1 Introduction

Connected Autonomous Vehicles (CAVs) are a transformative innovation in the transportation industry, combining autonomous driving capabilities with advanced connectivity features. These vehicles can communicate with each other, infrastructure, and other road users to enhance safety, efficiency, and convenience. The integration of technologies such as artificial intelligence (AI), sensors, and vehicle-to-everything (V2X) communication is driving the development and deployment of CAVs, promising a future where transportation is smarter and more sustainable.

3.2 Reason for Emergence

I. Safety Concerns

Human error is a leading cause of traffic accidents, and it can lead to fatalities. CAVs aim to reduce accidents by eliminating human error through advanced driver-assistance systems (ADAS) and autonomous driving technologies.

II. Technological Advancements

Rapid progress in AI, machine learning, and sensor technologies has made it possible to develop vehicles that can navigate complex environments autonomously. Improvements in connectivity infrastructure, such as 5G networks, enable real-time data exchange between vehicles and their surroundings.

III. Environmental Impact

CAVs can contribute to reducing emissions by optimizing driving patterns and integrating with electric vehicle (EV) technology. This aligns with global efforts to combat climate change and promote sustainable transportation.

3.3 Shaping Transportation Industry

I. Safety Improvement

CAVs are designed to reduce human error, which is a leading cause of traffic accidents. By using sensors, cameras, and AI, these vehicles can detect and respond to hazards more quickly and accurately than human drivers.

For example, advanced driver-assistance systems (ADAS) in CAVs can prevent collisions by automatically braking or steering away from obstacles and Waymo's autonomous vehicles have demonstrated a lower accident rate compared to human-driven vehicles.

II. Traffic Efficiency

By communicating with each other and traffic management systems, CAVs can optimize traffic flow, reduce congestion, and improve overall transportation efficiency. This can lead to shorter travel times and reduced fuel consumption.

The ideal scenario as all vehicles to move simultaneously when a traffic light turns green would be possible with the implementation of CAVs. All vehicles at an intersection are connected, the traffic light system can optimize the green light duration to ensure that all vehicles can move simultaneously and smoothly.

III. Enhance Accessibility

CAVs can provide greater mobility options for individuals who are unable to drive, such as the elderly or disabled. Autonomous ride-hailing services can offer convenient and affordable transportation solutions, enhancing independence and quality of life.

3.4 Summary

By addressing safety, efficiency, environmental impact, and accessibility, Connected Autonomous Vehicles (CAVs) are poised to revolutionize the transportation industry, making it safer, more efficient, and more sustainable.

Section 4: Screenshot of the webpage

4.1 Home Page



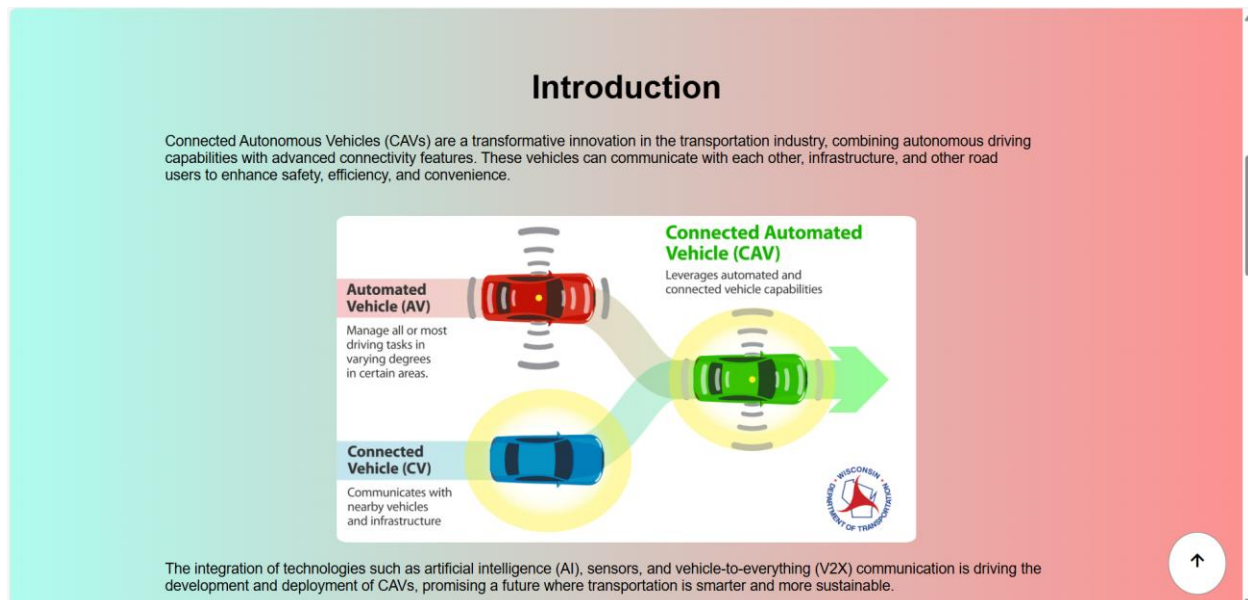
4.2 Content Menu (Burger Menu)



4.3 EXPLORE after hovering (Navigation button)



4.4 Introduction Section



4.5 Reason for Emergence Section



4.6 Breadcrumb Slider Button



4.7 Shaping Transportation Section

Shaping Transportation Industry



Safety Improvement

By using sensors, cameras, and AI, these vehicles can detect and respond to hazards more quickly and accurately than human drivers.



Traffic Efficiency

By communicating with each other and traffic management systems, CAVs can optimize traffic flow, reduce congestion, and improve overall transportation efficiency.



Enhance Accessibility

CAVs can provide greater mobility options for individuals who are unable to drive, such as the elderly or disabled.

↑

4.8 Summary, The Team & Footer Section

Summary

By addressing safety, efficiency, environmental impact, and accessibility, Connected Autonomous Vehicles (CAVs) are poised to revolutionize the transportation industry, making it safer, more efficient, and more sustainable.

Fact

The Team

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© Copyright 2025 B240253C | Site designed by Calvin

↑

4.9 Fact & Hide Button and Hyperlink in Summary Section

Summary

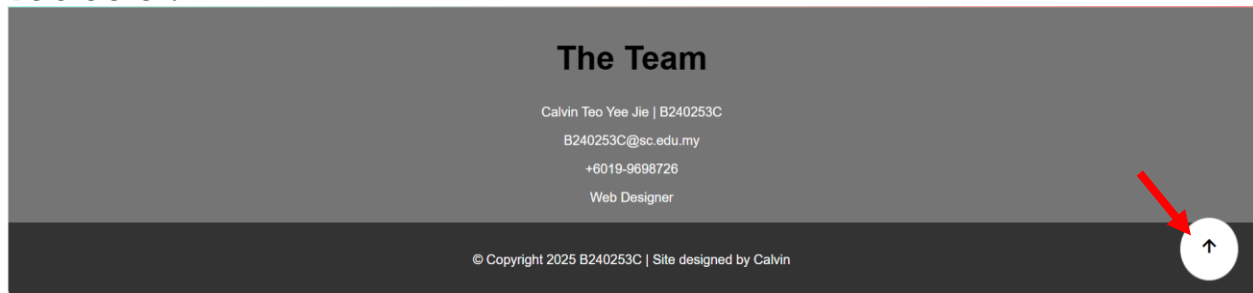
By addressing safety, efficiency, environmental impact, and accessibility, Connected Autonomous Vehicles (CAVs) are poised to revolutionize the transportation industry, making it safer, more efficient, and more sustainable.

Hide

A total of around 600,000 road accident cases have been recorded throughout 2023, and in that time 6,443 lives were lost to road accidents, which translates to an average of 18 deaths every day last year. Of these casualties, 65% were motorcyclists, comprised of both riders and pillion riders, aged 16 to 35 years.
[Visit to read more.](#)

4.10 Go to Top Button

Before click:



After click: Navigate to the top of website



Section 5: Conclusion

I implemented a responsive layout using HTML and CSS to create a structured content and some styling. JavaScript brought the site to life, allowing interactive elements such as drop-down menus and slider to enhance user experience. I also integrated multimedia elements like images, icons, and button to make the information more engaging and easier to understand.

This assignment has been a rewarding experience as Connected Autonomous Vehicles (CAVs) is one of my interests since I was young. Through my research, I gained insights into how autonomous vehicles operate, the role of connectivity in improving traffic safety and efficiency, and the challenges the industry still faces—such as ethical concerns, cybersecurity risks, and infrastructure readiness.

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